
EM 602: Experiments in Materials Science

Course Code	EM 602
Course Name	Experiments in Materials Science
L-T-P-C	2-0-3-4
Intended for	M.Tech Energy Materials
Distribution	Core course (Semester I)
One-time approval	recommended by the 14th BoA (Sep 2016)

Course Modules

- **Module 1 (Growth of Nanostructures and Thin films):** Nucleation, Crystal growth, Growth mechanism and kinetics of crystal facets, Size and shape controlled synthesis of nanomaterials. Materials synthesis from solution and melt, Growth of thin films, Thin film deposition by thermal evaporation. (9)
- **Module 2 (Structural and Microstructural Characterization):** Experimental exploration of energetics and structure of materials, demonstration to quantify energy (DSC), structure (X-ray diffraction, electron diffraction) and chemical composition (EDAX) in condensed matter. Principles and applications of imaging techniques for materials characterization, scanning electron microscopy (SEM), imaging modes and detectors. electron-specimen interactions, interaction volume. Transmission electron microscopy (TEM), elastic and inelastic scattering, modes of operation, imaging defects and strain fields, phase contrast imaging, Moire fringes, Fresnel fringes and high-resolution imaging. (10)
- **Module 3 (Properties):** Structure-property relationship through practical examples. Mechanical testing to investigate materials strength Electrical transport measurements, cyclic voltammetry. (9)

Text Books

1. J. Goldstein et al. *Scanning Electron Microscopy and X-Ray Microanalysis: A Textbook for Biologists, Materials Scientists and Geologists*, Springer 2003.
2. A.R. West, *Solid State Chemistry and Its Applications*, Wiley, 1985

References

1. Barry Carter, *Transmission Electron Microscopy: A text book for materials science*, Springer, 2009
2. B.D. Cullity, *Elements of X-Ray Diffraction*, Prentice Hall, 2001
3. C.R. Brundle, C.A. Evans and S. Wilson, *Encyclopedia of Materials Characterization*, Butterworth-Heinemann 1992